HOW AN LLM PROCESSES YOUR PROMPT

From Input to Output — End-to-End

A step-by-step trace of one fixed prompt through every stage of the pipeline

Our fixed prompt throughout this deck:

"Explain black holes simply"

THE PIPELINE

1	2	3	4	5	6	7	8	9	10
User Input	Prompt Engineering	Tokenization	Embedding	Transformer	Token Prediction	Decoding	Output Generation	Post-processing	Final Output

Language → Math

Prompts are converted to numbers (vectors) before the model ever sees them

Context is Everything

The full prompt is processed simultaneously, not word by word

Probabilistic Output

Every response is a sequence of statistically predicted tokens

STEP 1

Raw User Input

User types:

```
"Explain black holes simply"
```

What the user has

- A natural language question
- Vague intent — no structure
- No format specified
- No audience specified

What the AI needs

- Clear task definition
- Target audience context
- Output format guidance
- Scope constraints

The gap (problem)

- Raw input is ambiguous
- Model must guess everything
- Results will be inconsistent
- This is why PE matters

The prompt is just raw text — it has zero structure until the PE layer processes it.

STEP 2

Prompt Engineering Layer

User Input — **Prompt Eng** — Tokenize — Embed — Transformer — Predict — Decode — Generate — Post-proc — Output

BEFORE (raw)

"Explain black holes simply"

PE Layer

AFTER (engineered)

System: You are a science teacher.
Task: Explain black holes simply. Use an analogy.

1 Role

You are a science teacher.

2 Task

Explain black holes simply.

3 Constraint

Use an analogy. Keep it short.

4 Format

One paragraph, plain English.

The PE layer converts vague human intent into a precise, structured instruction the model can act on reliably.

STEP 3

Tokenization

User Input — Prompt Eng — **Tokenize** — Embed — Transformer — Predict — Decode — Generate — Post-proc — Output

"You are a science teacher. Explain black holes simply. Use an analogy."

AFTER TOKENIZATION — Each unit the model reads:

You	are	a	science	teacher	.	Explain	black
ID: 1831	ID: 526	ID: 257	ID: 3783	ID: 4700	ID: 13	ID: 48366	ID: 2042

holes	simply	.	Use	an	analogy	.
ID: 9441	ID: 5752	ID: 13	ID: 5765	ID: 281	ID: 26890	ID: 13

Key fact: Tokens are not always full words. A token ≈ 4 characters on average. Numbers, punctuation, and prefixes get their own tokens. This prompt = 15 tokens.

STEP 4

Embedding (Token → Vector)

EACH TOKEN IS LOOKED UP IN A LEARNED EMBEDDING MATRIX:

black	[	0.82	-0.41	0.63	0.29	-0.75	0.18	0.93	-0.55 ...]
holes	[	-0.31	0.77	0.55	-0.22	0.66	-0.48	0.12	0.84 ...]
simply	[	0.45	0.33	-0.29	0.71	0.22	0.90	-0.63	0.38 ...]

768 to 12,288 dims — Real embeddings have hundreds to thousands of dimensions, not just 8 shown here

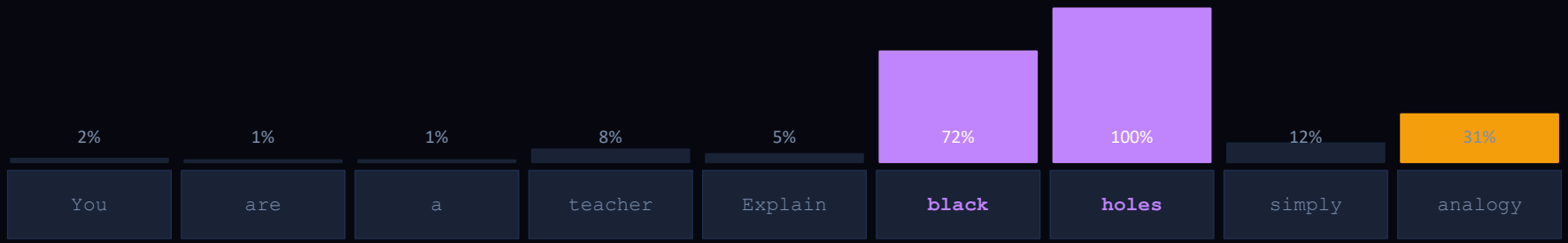
Semantic meaning — Similar words cluster nearby in vector space — 'black holes' and 'gravity' are close

Pre-trained lookup — Embeddings come from training, not computed fresh. Each token has a fixed starting vector

STEP 5

Transformer — Attention & Context

ATTENTION WEIGHTS — 'holes' attends to these tokens most strongly:



Multi-Head Attention

"holes" attends strongly to "black" (72%) and "analogy" (31%). Irrelevant tokens near-zero. Runs in parallel across all tokens at once.

Feed-Forward Layers

After attention, each token passes through neural layers that apply learned patterns — associating "black holes" with physics concepts.

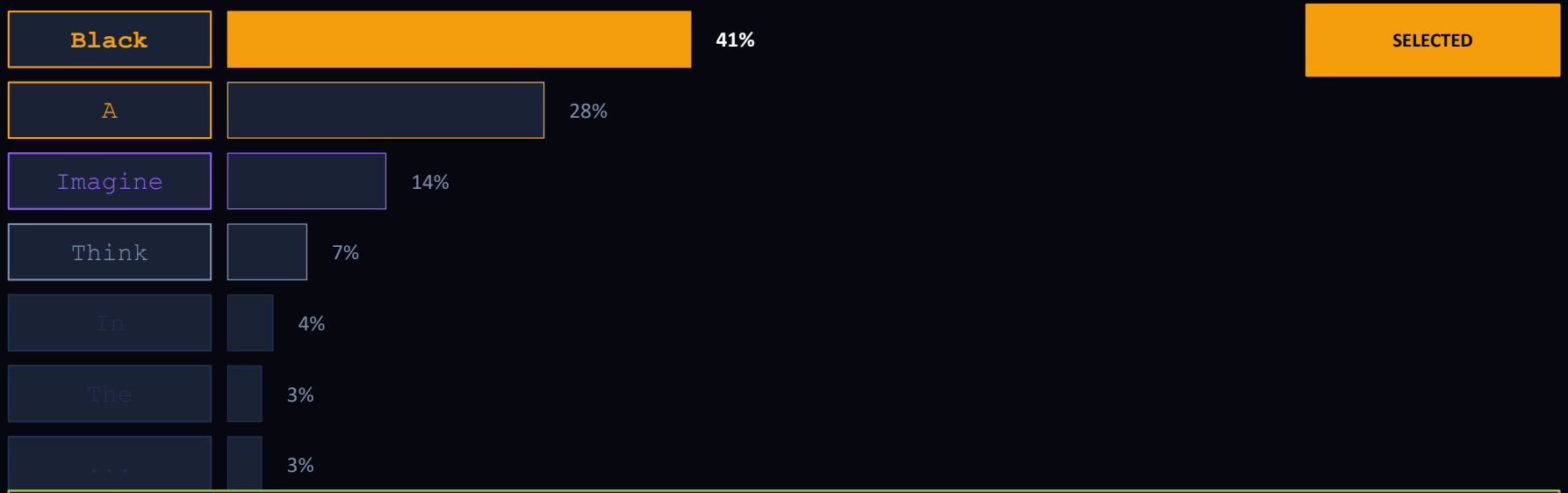
96+ Stacked Layers

GPT-4 class models have 96 layers. Each layer refines the understanding. Early layers = syntax, deep layers = reasoning and semantics.

STEP 6

Probabilistic Token Prediction

PREDICTING THE FIRST OUTPUT TOKEN — Model assigns a probability to every word in its 50,000+ vocabulary:



This happens for EVERY token in the response. "Black" wins this round. Now the model predicts the next token given "Black"... and repeats until done.

STEP 7

Decoding Strategy

DECODING CONTROLS HOW THE MODEL PICKS FROM ITS PROBABILITY DISTRIBUTION:

Greedy Decoding

`Temp = 0`

Always picks the highest-probability token. Fast, deterministic, but can be repetitive.

Sample output:

"Black holes are regions of space..."

Temperature Sampling

`Temp = 0.7`

Introduces randomness. Slightly lower-ranked tokens can win. Balanced creativity vs accuracy.

Sample output:

"Black holes are cosmic objects..."

High Temperature

`Temp = 1.5`

More randomness. Model picks unexpected tokens more often. Creative but risks incoherence.

Sample output:

"Black holes swallow the universe..."

Top-P (Nucleus)

`p = 0.9`

Only samples from tokens summing to top 90% of probability. Eliminates tail-end nonsense tokens.

Sample output:

"Imagine a cosmic drain in space..."

STEP 8

Token-by-Token Output Generation

User Input — Prompt Eng — Tokenize — Embed — Transformer — Predict — Decode — **Generate** — Post-proc — Output

AUTOREGRESSIVE GENERATION — each new token feeds back into the model as input:

Pass 1	Context: "[prompt]"	"Black"
Pass 2	Context: "[prompt] + Black"	"holes"
Pass 3	Context: "[prompt] + Black holes"	"are"
Pass 4	Context: "[prompt] + Black holes are"	"regions"
Pass 5	Context: "[prompt] + Black holes are regions"	"of"

Accumulating: "Black holes are regions of space where gravity is so strong that nothing — not even light — can escape..."

STEP 9

Post-processing Layer

RAW OUTPUT FROM MODEL — before post-processing:

```
"black holes are regions of space where gravity is so strong that not even light can escape. think of it like
a really powerful vacuum cleaner but in space <end_token>"
```

A Formatting

Capitalise sentences. Remove special tokens like <end_token>. Apply requested format (paragraph, bullets, JSON). Add spacing.

B Safety Filtering

Check output against harm policies. Flag or suppress content that violates guidelines. This happens before the user sees anything.

C Output Validation

In production systems, verify the response is complete. Check for truncation. Validate JSON if structured output was requested.

AFTER POST-PROCESSING:

```
"Black holes are regions of space where gravity is so strong that not even light can
escape. Think of it like a really powerful vacuum cleaner — but in space."
```

STEP 10

Final Output — Delivered to User

Final output delivered to user:

"Black holes are regions of space where gravity is so strong that not even light can escape. Think of it like a really powerful vacuum cleaner — but in space. Once something crosses the boundary called the event horizon, it cannot come back out."

COMPLETE PIPELINE — ONE PROMPT, NINE TRANSFORMATIONS:

1	2	3	4	5	6	7	8	9	10
Raw Input	PE Layer	Tokenize	Embed	Transformer	Predict	Decode	Generate	Post-proc	Output
"Explain black holes simply"	Role + Task + Constraint added	15 tokens assigned numeric IDs	IDs → 768-dim float vectors	Attention + 96 neural layers	Probability over 50k+ vocab	Temp=0.7 sampling strategy	Token-by-token autoregressive	Format + safety + validate	Clean text → user